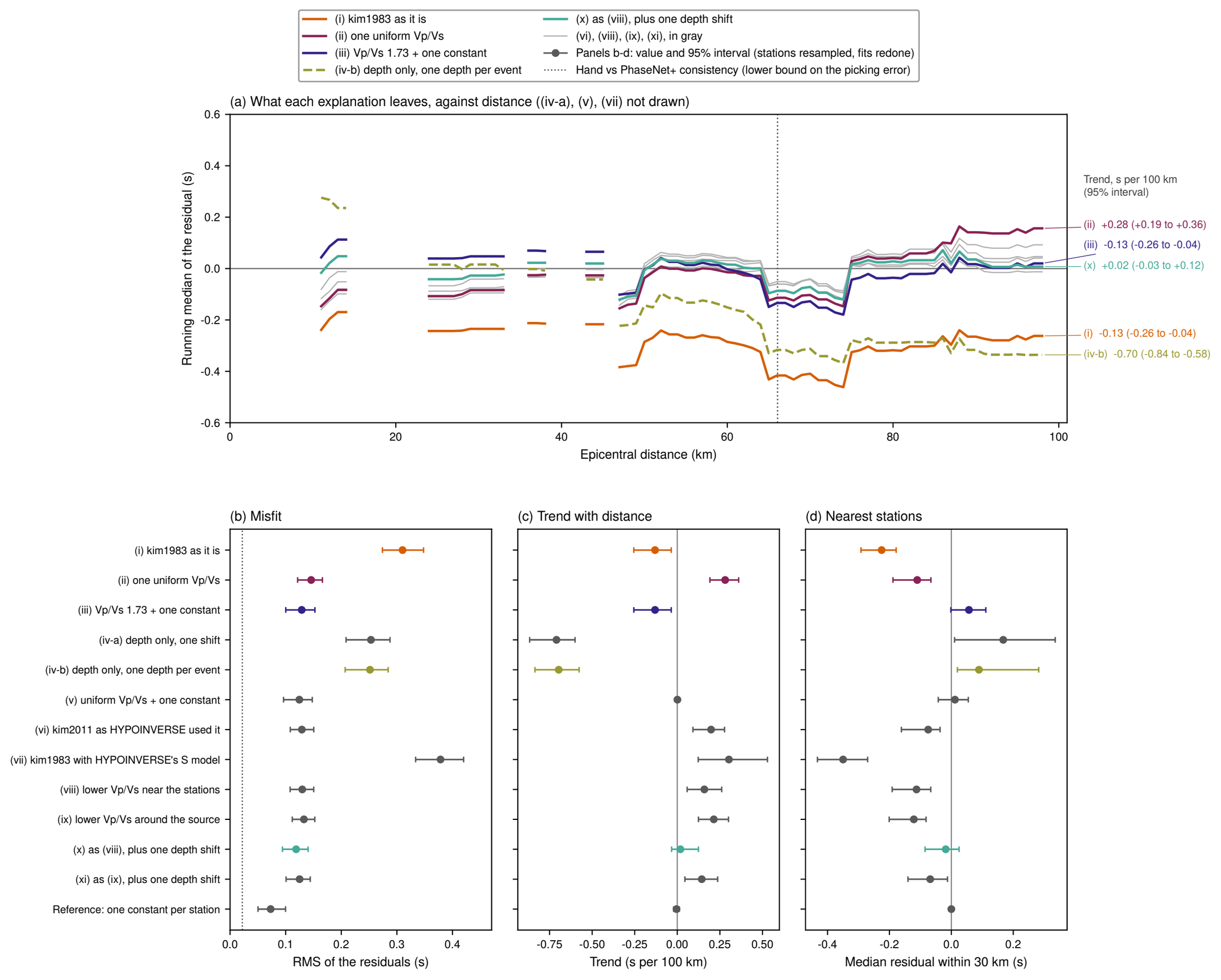
(i) kim1983 as it is
(ii) one uniform Vp/Vs
(iii) Vp/Vs 1.73 + one constant
(iv-b) depth only, one depth per event
(x) as (viii), plus one depth shift
(vi), (viii), (ix), (xi), in gray
Panels b-d: value and 95% interval (stations resampled, fits redone)
Hand vs PhaseNet+ consistency (lower bound on the picking error)
(a) What each explanation leaves, against distance ((iv-a), (v), (vii) not drawn)
Running median of the residual (s)
Epicentral distance (km)
Trend, s per 100 km
(95% interval)
(ii) +0.28 (+0.19 to +0.36)
(iii) -0.13 (-0.26 to -0.04)
(x) +0.02 (-0.03 to +0.12)
(i) -0.13 (-0.26 to -0.04)
(iv-b) -0.70 (-0.84 to -0.58)
(b) Misfit
(c) Trend with distance
(d) Nearest stations
(i) kim1983 as it is
(ii) one uniform Vp/Vs
(iii) Vp/Vs 1.73 + one constant
(iv-a) depth only, one shift
(iv-b) depth only, one depth per event
(v) uniform Vp/Vs + one constant
(vi) kim2011 as HYPOINVERSE used it
(vii) kim1983 with HYPOINVERSE's S model
(viii) lower Vp/Vs near the stations
(ix) lower Vp/Vs around the source
(x) as (viii), plus one depth shift
(xi) as (ix), plus one depth shift
Reference: one constant per station
RMS of the residuals (s)
Trend (s per 100 km)
Median residual within 30 km (s)